1. PRODUCT & COMPANY IDENTIFICATION

Product Name	: Hitary 928 Poly Putty
Product Code	: HYPU-928
Intended Use	: Body filler for professional use only.
Manufacturer	: Mercury Paints Factory Sdn Bhd
Malaysia Address	: Lot 10, Jalan Perusahaan4, Batu Caves Industrial Estate, 68100 Selangor, Malaysia.
Telephone no.	: +60361887005 (Marketing Dept)
Facsimile no.	: +60361886081

2. POSSIBLE HAZARDS

Hazard Pictogram:

Signal word: Danger

This preparation is hazardous per the following P.U. (A) 310 Schedule (Regulation 2013)

Flammable liquids	Category 3
Skin corrosion or irritation	Category 2
Serious eye damage or eye irritation	Category 2
Specific target organ toxicity-repeated exposure	Category 1
Aspiration hazard	Category 1
Acute aquatic toxicity	Category 2
Health hazards: Acute toxicity (inhalation)	Category 4

Hazard Statement

1) H226 - Flammable liquid and vapour
2) H315 - Causes skin irritation
3) H319 - Causes serious eye irritation
4) H372 - Causes damage to organs through prolonged or repeated exposure
5) H304 - May be fatal if swallowed and enters airways.
6) H335 - May cause respiratory irritation
7) H401 - Toxic to aquatic life
8) H332 - Harmful if inhaled

Health Hazards

1) Inhalation - May cause drowsiness and irritation of respiratory tract. Avoid inhaling vapour.
2) Skin - May cause irritation on prolonged contact, which can cause dermatitis.
3) Eyes - Irritation and soreness
4) Ingestion - Sore throat, stomachache, nausea

Precautionary statements:

Prevention

1) P210 - Keep away from heat/sparks/open flames/hot surfaces. - No smoking.
2) P233 - Keep container tightly closed.
3) P241 - Use explosion-proof electrical, ventilating, lighting and all material-handling equipment.
4) P242 - Use only non-sparking tools
5) P243 - Take precautionary measures against static discharge.
6) P260 - Do not breathe dust/fume/gas/mist/vapours/spray
7) P261 - Avoid breathing vapours.
8) P264 - Wash hands thoroughly after handling.
9) P270 - Do not eat, drink or smoke when using this product.
10) P271 - Use only outdoors or in a well-ventilated area.
11) P273 - Avoid release to the environment
12) P280 - Wear protective gloves/ eye protection/ face protection.

Response

1) P302 + P352 - IF ON SKIN: Wash with plenty of soap and water.
2) P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.
3) P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing..
4) P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses. If present and easy to do. Continue rinsing.
5) P301+P330+P331: IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
6) P312 Call a doctor/physician if you feel unwell.
7) P331 Do not induce vomiting.
8) P332 + P313 If skin irritation occurs, get medical advice.
9) P337 + P313 If eye irritation persists, get medical advice.
10) P370 + P378 In case of fire. Use suitable extinguishing media for extinction (refer section 5).

Storage

1) P403 + P233 Store in a well-ventilated place. Keep container tightly closed.
2) P403 + P235 Store in a well-ventilated place. Keep cool.
3) P405 Store locked up.

Disposal

1) P501 Dispose of contents/container in accordance with local/regional/national/international regulation

3. COMPOSITION / INFORMATION OF THE INGREDIENTS

Chemical composition contains synthetic resins, filler and diluent.

Substances that present a hazard to health and the environment found in this product.
For more information on this hazardous substance, please refer below.

Hazardous components:

Chemical Name	CAS No.	%
Styrene	100-42-5	< 35
Cobalt-(II)-2-ethylhexanoate	136-52-7	< 0.5

4. FIRST-AID MEASURES

General advice	: When symptoms persist or in all cases of doubt seek medical advice. Never give anything by mouth to an unconscious person. : Take appropriate steps to avoid fire, explosion and inhalation hazards. DO NOT DELAY.
Inhalation	: When exposed to large amounts of steam and mist, move to fresh air. : Take specific treatment if needed. : Get medical attention immediately. : If breathing is stopped or irregular, give artificial respiration and supply oxygen.
Skin contact	: Flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. : Wash contaminated clothing thoroughly before re-using. : Get medical attention immediately. : Go to the hospital immediately if symptoms (flare, irritate) occur. : Wash thoroughly after handling.
Eye contact	: Do not rub your eyes. : Immediately flush eyes with plenty of water for at least 15 minutes and call a doctor/physician. : Get medical attention immediately. : Go to the hospital immediately if symptoms (flare, irritate) occur. : Remove contact lenses if worn.
Ingestion	: Please be advised by doctor whether induction of vomit is demanded or not. : Rinse your mouth with water immediately. : Get medical attention immediately. : If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Stop if the exposed person feels sick as vomiting may be dangerous. : Maintain an open airway. Loosen tight clothing.
Most important symptoms and effects, both acute and delayed	: Breathing of high vapour concentrations may cause central nervous system (CNS) depression resulting in dizziness, light-headedness, headache, nausea and loss of coordination. Continued inhalation may result in unconsciousness and death. Eye irritation signs and symptoms may include a burning sensation, redness, swelling, and/or blurred vision. Respiratory irritation signs and symptoms may include a temporary burning sensation of the nose and throat, coughing, and/or difficulty breathing. Skin irritation signs and symptoms may include a burning sensation, redness, swelling, and/or blisters. Auditory system effects may include temporary hearing loss and/or ringing in the ears. Visual system disturbances may be evidenced by decreases in the ability to discriminate between colours.
Immediate medical attention, special treatment	: Potential for chemical pneumonitis. Call a doctor or poison control centre for guidance.

5. FIRE FIGHTING MEASURES

Extinguishing media	: Carbon Dioxide / dry powder / alcohol-resistant foam / universal aqueous film-forming foam. DO NOT use water jets.
Extinguishing media which shall not be used for safety reasons	: High volumn water jet
Hazardous combustion	: Burning materials emits toxic fumes and smoke avoid inhalation of fumes of burning products. Fire will produce dense black smoke containing hazardous combustion products. Exposure to decomposition products may be a hazard to health.
Protection of fire fighters	: Wear breathing apparatus and protective firefighting clothing. : Do not allow run-off from firefighting to enter drains or water courses. : Cool containers with water until well after fire is out. : Avoid inhalation of materials or combustion by-products. : Use appropriate extinguishing measure suitable for surrounding fire. : Vapour or gas is burned at distant ignition sources can be spread quickly. : Due to the extremely low flash point, irrigating fire extinguishing may be less effective when put out a fire.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	: Keep in a well-ventilated place. : Do not breath vapours. : Ventilate closed spaces before entering. : Must work against the wind, let the upwind people to evacuate. : Do not touch spilled material. Stop leak if you can do it without risk. : Remove all sources of ignition. : Do not direct water at spill or source of leak. : Avoid skin contact and inhalation. : Cleanup and disposal under expert supervision is advised. : Keep unauthorized people away, isolate hazard area and deny entry
Environmental precautions	: Notify the respective authorities in accordance with local law in the case of contamination of rivers, lakes or waste water systems. : Prevent runoff and contact with waterways, drains or sewers.
Methods and materials for containment and cleaning up	: Contain and collect spillage with non-combustible absorbent materials. e.g sand, earth, vermiculite diatomaceous earth and place in container for disposal according to local regulations. Clean preferably with a detergent; avoid use of solvents. : Large portion spill-out: stay upwind and keep out of low areas. Dike for later disposal.

: Notify the central and local government if the emission reach the standard threshold.
: Appropriate container for disposal of spilled material collected.
: Small leak: sand or other non-combustible material, please let use absorption.
: Wipe off the solvent
: Dike for later disposal.
: Do not use plastic containers.

7. HANDLING AND STORAGE

Safe Handling	: Avoid prolonged or repeated contact with skin. Extinguish any naked flames. Do not empty into drains. After use lid must be closed air-tight. Use only in well ventilated areas. : Dealing only with a well-ventilated place. : Do not inhale the steam prolonged or repeated. : Avoid contact with heat, sparks, flame or other ignition sources. : Contaminated work clothing should not be allowed out of the workplace. : The vapour is heavier than air. Beware of accumulation in pits and confined spaces. Use local exhaust ventilation if there is risk of inhalation of vapours, mists or aerosols. Bulk storage tanks should be diked (bunded).
Avoidance on Contact	: Strong oxidising agents. Copper alloys.
Handling Temperatures	: 10°C - 35°C
Safe Storage	: Store in accordance with the `Highly Flammable Liquids and Liquefied Petroleum Gases Regulations'. Keep away from direct sunlight and other sources of heat or ignition. Do not smoke in storage areas. : Keep in the original container : Keep sealed when not in use. : No open fire. : Prevent static electricity and keep away from combustible materials or heat sources : Collected them in sealed containers.
Storage Temperatures	: 10°C - 35°C
Other data	: Packaging material : Suitable material: For container paints, use epoxy paint, zinc silicate paint., For containers, or container linings use mild steel, stainless steel. Unsuitable material: Copper, Copper alloys.

8. EXPOSURE CONTROLS AND PERSONAL PROTECTION

Occupational Exposure Limits

	Type	mg/m3	ppm
OSHA	TWA	85.2	-
ACGIH	TWA	-	20
ACGIH	STEL	-	40

Engineering Control : Local Exhaust ventilation is recommended.
: Eye washes and shower for emergency use.
: Use only in well ventilated area.

Personal Protection

Ventilation : Use in well ventilated area with local exhaust ventilation. Ventilation required and equipment must be explosion proof. Use away from all ignition sources.

Respiratory Protection : Suitable respiratory protective equipment / safety mask must be used if facing concentrations above the exposure limit.

Eye Protection : Use safety eyewear.

Skin Protection : Impervious gloved must be worn. If contact is likely oil impervious clothing must be worn.

Hand Protection : The breakthrough time of gloves is unknown for the product itself. The glove given is recommended on basis of the substances in the preparation. The protective glove should be checked in each case for their work specific suitability, mechanical stability, product compatibility, and anti-static properties.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance / Form	: Medium high viscous paste.
Colour	: Grey or Brown
Odour	: Solvent
pH	: Not applicable
Melting point	: Not applicable
Boiling point	: Not applicable
Change in physical state	: Air tight – paste form. Solid when mixed with hardener.
Flash point	: 31°C
Specific gravity	: 1.35 – 1.55 g/ml
Evaporation rate	: Not available
Flammability (solid, gas)	: Not available
Vapour pressure	: 0.20 – 0.30kpa
Solubility in water	: Insoluble
Auto-ignition temperature	: 300 – 350°C
Decomposition temperature	: Not applicable
Viscosity	: Dynamic 1300 – 1800 mPa

10. STABILITY AND REACTIVITY

Stability	: Stable under normal use conditions. Prolong exposure to heat may be sufficient to raise the temperature above the product flash point. Thermal decomposition can give rise to acid fumes.
Conditions to Avoid	: Heat, flames and sparks.
	: Accumulation of electrostatic charges, Heating, Flames and hot surfaces
Material to Avoid	: Strong oxidizing agents.
Hazardous Decomposition Products	: Not applicable
SADT	: Not available

11. TOXICOLOGICAL INFORMATION

Information on toxicological effects

Basis for Assessment	: With reference to styrene as the hazardous substance.
Likely Routes of Exposure	: Inhalation, eye, skin is the primary route of exposure followed by accidental ingestion.

Hazardous ingredients:

Acute toxicity

Styrene	: LC50 Inhalation Gas, Rat, Dose 2770ppm, exposure 4 hours.
	: LC50 Inhalation Vapour, Rat, Dose 11800mg/m³, exposure 4 hours.
	: LD50 Oral, Rat, 2650mg/kg.
	: Dermal toxicity, expected to be of low toxicity: LD50 >5000 mg/kg
Cobalt-(II)-2-ethylhexanoate	: LD50 Dermal, Rabbit, Dose >5g/kg
	: LD50 Oral, Rat, 3129mg/kg

Irritation

Styrene	: Eyes - mild irritant, Human, Exposure 50ppm
	: Eyes - moderate irritant, Rabbit, Exposure 24hours 100mg
Cobalt-(II)-2-ethylhexanoate	: Irritating

Serious Eye Damage/Irritation

Styrene : Causes serious eye irritation

Respiratory Irritation

Styrene : Inhalation of vapours or mists may cause irritation to the respiratory system.

Respiratory or Skin Sensitization

Styrene : Not expected to be a sensitizer

Allergic Sensitization

Cobalt-(II)-2-ethylhexanoate : Skin – may cause an allergic skin reaction

Specific Target organ toxicity (single exposure)

Styrene : Exposure routes – inhalation
: Respiratory system – may cause respiratory irritation.
: Category 3

Specific Target organ toxicity (repeated exposure)

Styrene : Exposure routes – inhalation
: Respiratory system – may cause damage to organ through prolonged or repeated exposure.
: Category 1
: Harmful: danger of serious damage to health by prolonged exposure through inhalation. Can cause liver damage. Auditory system: prolonged and repeated exposures to high concentrations have resulted in hearing loss in rats. Solvent abuse and noise interaction in the work environment may cause hearing loss. Central nervous system: repeated exposure affects the nervous system.

Germ Cell Mutagenicity : Not considered a mutagenic hazard

Carcinogenicity : Not expected to be carcinogenic. Styrene has been found to produce lung tumours in mice. These tumours are not considered to be relevant to human.

Carcinogenicity Classification

Styrene : ACGIH Group A4: Not classifiable as a human carcinogen

Reproductive toxicity

Styrene : Suspected of damaging fertility or the unborn child.
Cobalt-(II)-2-ethylhexanoate : May damage fertility or the unborn child.

Further information:

Potential acute health effects

Eye contact : Causes serious eye irritation.
Inhalation : Harmful if inhaled.
Skin contact : Causes skin irritation. May cause an allergic skin reaction.
Ingestion : No known significant effects or critical hazards.

Symptoms / routes of exposure

Exposure to component solvent vapour concentrations in excess of the exposure limit may result in adverse health effect.

Inhalation : Vapour may be irritating to the eyes and the respiratory tract, may cause headache, muscular weakness, dizziness, drowsiness.
Skin contact : Low order of toxicity. Frequent or prolonged contact may dry the skin, leading to discomfort.
Eye contact : Will cause eye discomfort, may injure eye tissue if not remove promptly.

12. ECOLOGICAL INFORMATION

Ecotoxicity

Styrene	: Fish, LC50 10mg/ℓ 96 hr Pimephales promelas (OECD TG 203. GLP)(ECHA) : Crustaceans, EC50 4.7 mg/ℓ 48 hr Daphnia magna (OECD TG 202, GLP)(ECHA) : Algae, EC50 4.9 mg/ℓ 72 hr Selenastrum capricornutum (EPA OTS 797.1050, GLP)

Persistence and degradability

Persistence

Styrene	: log Kow 2.96 log Kow (OECD TG 107)(ECHA)

Degradability

Styrene	: Not available

Bioaccumulative potential

Bioaccumulative potential	: Not available

Biodegradation

Styrene	: Biodegradability = 100 (%) (existing chemical safety inspections data)

Mobility in soil	: Not available
Other adverse effect	: Not available

13. DISPOSAL CONSIDERATIONS

The material and any contaminated container should be disposed of in accordance with the Environmental Act.

Waste & Material Disposal	: Recover or recycle if possible. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste classification and disposal methods in compliance with applicable regulations. Do not dispose into the environment, in drains or in water courses. Waste product should not be allowed to contaminate soil or water.
Container Disposal	: Drain container thoroughly. After draining, vent in a safe place away from sparks and fire. Send to drum re-coverer or metal reclaimer. Residues may cause an explosion hazard. Do not puncture, cut or weld uncleaned drums.
Legal Legislation	: Disposal should be in accordance with applicable regional, national, and local laws and regulations.

14. TRANSPORT INFORMATION

Transport only in accordance with the requirements of the Carriage of Dangerous Goods by Road and Rail (Classification, Packaging and Labeling), ADR for road, RID for rail, IMDG for sea and ICAO/IATA for air transport.

Land, Sea and Air Transport

Shipping Name	: Paints related products
UN NO	: UN 1263
Class	: 3
Packing Goup	: III
Environment Hazard	: No

15. REGULATORY INFORMATION

Classification according to the Malaysia Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Act 1994, Regulations 2013, P.U. (A) 310.

Safety Phrases	: Do not breathe vapour/spray. In case of insufficient ventilation wear suitable respiratory equipment. Restricted for professional use only.

Reference should be made to the Environment Protection Act and its regulations.

16. OTHER INFORMATION

This product is intended for industrial and professional use only.

The information herein is extracted from the SDS of the various hazardous ingredients which we have received from our suppliers and is based on the present state of knowledge on such ingredients. The information given is only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. This SDS is given in good faith and the users' working conditions are beyond our knowledge and control. The product is not to be used for other purposes than those specified. It is always the responsibility of the party handling/using this product to take all the necessary steps in order to comply with the local laws and regulations. The information in this SDS is meant as a description of the safety requirements of the product and it is not to be considered as a guarantee or warranty of the product's properties, express or implied.

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